

AP Calculus — Trace Packet

Instructions: For each item below, copy the lines under "Copy this exactly" onto a fresh sheet of paper, by hand, in your own natural handwriting. Do not solve the problem yourself or change anything. One item per page. Photograph each page (well-lit, full page in frame, no name on the page). Save each photo as the Item ID shown (e.g. `CALC-E1.jpg`).

CALC-E1 (easy)

Problem (for context only — do not solve independently): Find $f'(x)$ for $f(x) = 4x^3 - 2x + 7$.

Copy this exactly, by hand:

$$f'(x) = 12x^2 - 2$$

CALC-E2 (easy)

Problem (for context only — do not solve independently): Evaluate the definite integral: the integral from $x=0$ to $x=2$ of $3x^2$ dx.

Copy this exactly, by hand:

$$\text{integral of } 3x^2 \text{ dx from } 0 \text{ to } 2 = [x^3] \text{ from } 0 \text{ to } 2 = 8$$

CALC-E3 (easy)

Problem (for context only — do not solve independently): A circle's area is $A = \pi r^2$. Write the related-rates equation for dA/dt in terms of r and dr/dt .

Copy this exactly, by hand:

$$dA/dt = 2 \pi r (dr/dt)$$

CALC-H1 (hard)

Problem (for context only — do not solve independently): Find $f'(x)$ for $f(x) = (2x + 1) / (x^2 + 3)$ using the quotient rule. Show your work.

Copy this exactly, by hand:

$$f'(x) = [2(x^2+3) - (2x+1)(2x)] / (x^2+3)^2$$

$$f'(x) = [2x^2 + 6 - 4x^2 - 2x] / (x^2+3)^2$$

$$f'(x) = (-2x^2 - 2x + 6) / (x^2+3)^2$$

CALC-H2 (hard)

Problem (for context only — do not solve independently): Find $g'(x)$ for $g(x) = x^2 * e^{(-3x)}$ using the product rule. Show your work.

Copy this exactly, by hand:

$$g'(x) = (d/dx[x^2]) e^{(-3x)} + x^2 (d/dx[e^{(-3x)}])$$

$$g'(x) = 2x e^{(-3x)} + x^2 (-3) e^{(-3x)}$$

$$g'(x) = e^{(-3x)} (2x - 3x^2)$$

CALC-H3 (hard)

Problem (for context only — do not solve independently): Evaluate the indefinite integral: the integral of $2x(x^2+1)^4 dx$.

Copy this exactly, by hand:

$$\text{let } u = x^2 + 1, \quad du = 2x \, dx$$

$$\text{integral of } u^4 \, du = u^5/5 + C$$

$$= (x^2+1)^5 / 5 + C$$

CALC-H4 (hard)

Problem (for context only — do not solve independently): An inverted conical tank has radius 3 m and height 6 m at the top. Water drains at $dV/dt = -2 \text{ m}^3/\text{min}$. Using similar triangles ($r = h/2$), find dh/dt when $h = 4 \text{ m}$.

Copy this exactly, by hand:

$$r = h/2 \text{ (similar triangles, } R=3, H=6)$$

$$V = (1/3) \pi r^2 h = (1/3) \pi (h/2)^2 h = (\pi h^3) / 12$$

$$dV/dt = (\pi h^2 / 4) (dh/dt)$$

$$-2 = (\pi (4)^2 / 4) (dh/dt) = 4 \pi (dh/dt)$$

$$dh/dt = -1/(2 \pi) \text{ m/min}$$

CALC-H5 (hard)

Problem (for context only — do not solve independently): Find the interval of convergence of the series: the sum from $n=1$ to infinity of $(x-2)^n / (n \cdot 3^n)$.

Copy this exactly, by hand:

ratio test: limit of $|a_{n+1}/a_n| = |x-2| / 3$

converges when $|x-2| < 3$, i.e. $-1 < x < 5$

at $x = -1$: sum of $(-1)^n / n$ converges (alternating harmonic)

at $x = 5$: sum of $1/n$ diverges (harmonic series)

interval of convergence: $[-1, 5)$